

PROFESSIONAL SUMMARY

Computer Science and Engineering graduate specializing in Machine Learning, Deep Learning, and Large Language Models. Proven experience in fine-tuning LLMs (BioGPT), developing real-time anomaly detection systems, and building deep learning solutions for medical image analysis. Oracle Certified Professional with proficiency in Python, PyTorch, and TensorFlow, focused on solving real-world problems using AI.

EDUCATION

•Bachelor of Technology in Computer Science and Engineering	2022-2026
B.P. Poddar Institute of Management & Technology, Kolkata	CGPA: 8.62
•Higher Secondary (Science – PCMB), WBCHSE	2022
Anandapur Higher Secondary School, West Medinipur	Percentage : 94.8%
•Secondary, WBSE	2020
Teghari High School (H.S.) , West Medinipur	Percentage : 90.85%

EXPERIENCE

•Large Language Models Internship	June - July 2025
National Institute of Technical Teachers' Training & Research (NITTTR), Kolkata	On Site
– Developed biomedical QA system by fine-tuning BioGPT with LoRA, optimizing performance and efficiency. – Accelerated model performance by 40% and reduced VRAM usage by 75% through strategic fine-tuning of BioGPT with LoRA, enabling deployment on consumer-grade GPUs for wider accessibility. – Modified model configurations to enable CPU compatibility by setting float32 precision in config.json for efficient deployment and broader hardware support.	
•Training & Internship part-time	January - June 2025
Sabudh Foundation & STPI	Online
– Completed a comprehensive Machine Learning module, applying supervised and unsupervised techniques for model development and evaluation – Gained hands-on experience in data science workflows using Dataiku, developing proficiency in data analysis, visualization, and platform integration.	

PERSONAL PROJECTS

•Real-Time Anomaly Detection and Alert System
A real-time deep learning system for detecting violent activities in CCTV video streams.
– Developed a MobileNetV2 + LSTM-based model to capture temporal patterns for accurate violence detection. – Lightweight and easily deployable solution (approx. model size 49 MB), suitable for real-time surveillance. – Technology Used: TensorFlow, Keras, MobileNetV2, LSTM, OpenCV, Flask.
•Attention-Guided Kidney Stone Detection from Ultrasound Images
A deep learning system for automated kidney stone detection from ultrasound images using attention-based computer vision.
– Developed an EfficientNetV2-S + CBAM deep learning model with automated ROI extraction, transfer learning, data augmentation, and two-stage fine-tuning for binary image classification. – Achieved 99.15% accuracy, 99.73% sensitivity, and 98.49% specificity on 1,413 test images with Grad-CAM explainability. – Technology Used: Python, PyTorch, EfficientNetV2-S, CBAM, OpenCV, Scikit-learn, NumPy, Grad-CAM.
•Bitcoin Price Forecasting & Trading Strategy Backtesting
Developed a time series forecasting model using a stacked LSTM network to predict 30-minute prices.
– Engineered an LSTM model to forecast Bitcoin prices, resulting in a backtested trading strategy with a 139.92% simulated ROI. – Technology Used : LSTM, Keras, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib.

TECHNICAL SKILLS AND INTERESTS

Languages: Java, Python, HTML, CSS
Libraries: Keras, OpenCV, Scikit-learn, Matplotlib, PyTorch
Tools: VScode, Git, Github, Jupyter notebook, LaTeX
Frameworks: TensorFlow, Flask, REST API, Streamlit
Cloud/Databases: Oracle DB, SQL, MySQL
Relevant Coursework: Data Structures and Algorithms, Operating Systems, OOP, DBMS, Cloud Computing, Software Engineering.
Areas of Interest: ML, DL, NLP, Computer Vision, LLMs, Backend Development, Artificial Intelligence, Data Analysis, Data Science.
Soft Skills: Patience, Attention to detail, Active listening, Goal orientation, Rapid learning

CERTIFICATIONS

• Oracle Cloud Infrastructure 2025 Certified Generative AI Professional, [Verify]	Aug 2025
• Oracle Cloud Infrastructure 2025 Certified Developer Professional, [Verify]	Aug 2025
• Oracle Cloud Infrastructure 2025 Certified Data Science Professional, [Verify]	Sept 2025
• Oracle Cloud Infrastructure 2025 Certified Networking Professional, [Verify]	Sept 2025
• Oracle AI Vector Search Certified Professional, [Verify]	Sept 2025

ACHIEVEMENTS & PUBLICATIONS

• IEEE Published Research Paper, Deep Learning-Based Automated Violence Detection and Identification System for Video Surveillance Applications	Feb 2026
---	----------